

Requirement Traceability

Each requirement from [REQUIREMENTS_CHECKLIST.md](#) with its implementation, evidence and status. Paths without a prefix are relative to `src/LearnHub/`; documentation paths are relative to `docs/`. Every row below was re-verified by reading the file named in the *Implementation* column and by running the project, not copied from an earlier revision.

Measured evidence on the `feature/railway-firebase-sqlite` branch: **211 automated tests pass with 0 failures** (`dotnet test LearnHub.sln -c Release`), **42 browser checks pass** (14 Playwright tests × 3 viewports) and the seeded catalogue contains **8 categories, 12 courses (11 published, 1 draft), 64 learning resources, 11 quizzes, 54 questions and 216 answer options**. The build is warning-free because CI passes `-p:TreatWarningsAsErrors=true`. The presentation site is live on Firebase Hosting at <https://learnhub-wapp.web.app>.

Summary

	Total	Pass	Ready, not yet performed
Mandatory	110	108	2
Optional	6	6	0

Ready, not yet performed: DEP-01 and DEP-02. Everything they depend on is built and verified — `Dockerfile`, `railway.json`, `.env.example` and `scripts/set-app-url.sh` — and the `container` CI job builds the image and runs it exactly as Railway does (Production, `PORT=8080`, TLS terminated in front, a volume at `/data`), including a restart that proves the SQLite file survives. The Railway CLI is installed but **not authenticated**, so no Railway deployment has been made and no Railway URL is claimed anywhere in this repository. The steps that finish both are in [DEPLOYMENT.md](#).

Coverage of the assignment's requirement areas

Each area the brief asks for, the requirement IDs that carry it, and the one piece of evidence that settles it. The full rows follow.

Requirement area	IDs	How it is met	Verification
Web-based learning functionality	GUEST-01–10, STU-01–13, ADM-01–11	Browsers work without a plug-in: course catalogue, lessons, progress and quizzes are ordinary HTML pages served by MVC	<code>tests/LearnHub.Tests/Integration/StudentJourneyTests.A_new_student_c</code>
Interlinked pages (working hyperlinks between pages)	UI-02, UI-04, TECH-11	Shared navigation, course cards, breadcrumbs, pagination and previous/next lesson links; routes declared in <code>Infrastructure/ApplicationBuilderExtensions.cs</code>	<code>NavigationCrawlTests</code> follows every internal link from each role's start page and fa
HTML5	TECH-06	Razor views use HTML5 doctype, <code><header></code> , <code><nav></code> , <code><main></code> , <code><footer></code> , <code><section></code> , <code><article></code> , <code><table></code> , <code><form></code> and HTML5 input types	<code>Views/Shared/_Layout.cshtml</code> ; <code>axe-core</code> scans in <code>tests/e2e/specs/acces</code>
CSS	TECH-07	<code>wwwroot/css/site.css</code> and <code>admin.css</code> (about 3,500 combined lines) restyle Bootstrap 5.3.8 through CSS custom properties and Grid/Flexbox	Both stylesheets are served and screenshotted by the browser tests
UI/UX	UI-01–04	One Interchange design system: consistent navigation, cards, dashboards, tables, forms, empty states and status messages	<code>tests/e2e/specs/</code> (public, student, admin) plus the screenshots published on the Fi
Database connectivity	TECH-03, TECH-04, DB-01–10	EF Core 10 with SQLite, one context, one migration, applied at start-up	<code>tests/LearnHub.Tests/Data/DatabaseModelTests.cs</code> ; the <code>container</code> CI job
CRUD	ADM-03–11, DB-09, STU-06	Create, read, update and delete through real forms for courses, categories, resources, quizzes, questions, enrolments, users and messages	<code>tests/LearnHub.Tests/Integration/AdminManagementTests.cs</code> (10 tests)
Registration	GUEST-08, AUTH-01, VAL-04	<code>AccountController.Register</code> with strong-password rules and a hashed password	<code>AuthenticationTests.Registration_creates_a_student_signs_them_in_and</code>

Requirement area	IDs	How it is met	Verification
Login / authentication	GUEST-09, AUTH-01-06, SEC-03	ASP.NET Core Identity cookie authentication, role claims, lockout and anti-forgery	<code>tests/LearnHub.Tests/Integration/AuthenticationTests.cs</code> (9 tests)
Registered-member functionality	STU-01-13	Dashboard, My Courses, lessons, progress, quizzes, results, profile, password change, logout	<code>StudentJourneyTests</code> (5 tests); <code>tests/e2e/specs/student.spec.js</code>
Administrator functionality	ADM-01-11	A protected Admin area with its own layout, dashboard and management pages	<code>tests/e2e/specs/admin.spec.js</code> ; <code>AuthorizationTests</code>
Database management (schema, constraints, migrations, seeds)	DB-01-10, TECH-04	Fluent API configuration, one migration creating 18 tables, idempotent seeding, <code>/health</code> database check	<code>DatabaseModelTests</code> ; <code>.github/workflows/ef-migrations.yml</code> ; <code>Infrastructure</code>
Form validation on the client	VAL-02, VAL-03	jQuery Validation Unobtrusive with custom adapters for the file and consent rules	<code>public.spec.js</code> contact and registration validation tests
Form validation on the server	VAL-01, VAL-04-06	DataAnnotations plus <code>ModelState</code> and service rules, so a tampered form is still rejected	<code>AuthenticationTests.Invalid_registration_is_rejected_by_server_side</code>
Client-side processing	TECH-08	Vanilla JavaScript and jQuery for the password meter, the quiz timer and counter, image previews and confirmation prompts	<code>public.spec.js</code> "registration form gives live password feedback"; <code>site.js</code>
Server-side processing	TECH-10, TECH-03	All routing, business rules, grading and persistence happen in C# controllers and services; no rule lives in the browser	<code>tests/LearnHub.Tests/Services/</code> (22 tests); <code>QuizAndCourseManagementTests.The_quiz_page_never_contains_the_correct</code>
Multimedia learning resources	TECH-09, GUEST-07, STU-08, ADM-05	Article, video (privacy-enhanced YouTube/Vimeo), PDF, image, link and exercise resource types, plus 12 SVG course covers and an interactive SVG map	<code>AdminManagementTests.Resources_are_created_validated_streamed_and_downloaded</code> ; <code>PublicSiteTests.Home_page_map_draws_one_line_per_category</code>
Functional navigation (menus, role-based links, protected routes, friendly 404)	UI-02, UI-04, AUTH-04, ERR-01	Role-aware navbar and admin sidebar, breadcrumbs, canonical redirects and a branded error page	<code>NavigationCrawlTests</code> (three roles); <code>PublicSiteTests.Unknown_pages_and_invalid_ids_return_a_friendly_404</code>
File organisation and naming	TECH-11, GIT-01	<code>src/</code> , <code>tests/</code> , <code>docs/</code> , <code>scripts/</code> , <code>FirebaseLanding/</code> ; consistent PascalCase types, one type per file, layered folders	Repository tree; ARCHITECTURE.md §2
.NET technologies	TECH-01-05, TECH-10	ASP.NET Core MVC, C#, Razor, EF Core 10, Identity — all on <code>net10.0</code>	<code>Directory.Build.props</code> ; the <code>build-and-test</code> CI job
Security	SEC-01-11, AUTH-02-06	Identity hashing, roles, anti-forgery, Razor encoding, CSP and security headers, IDOR and overposting guards, safe uploads, rate limiting	<code>tests/LearnHub.Tests/Unit/ContentSafetyTests.cs</code> , <code>FileStorageService</code>
Error handling	ERR-01-05	Friendly 404 and 500 pages, business-error messages, unique-constraint handling, structured logs	<code>PublicSiteTests.Unknown_pages_and_invalid_ids_return_a_friendly_404</code>
Responsiveness	UI-05	Bootstrap 5.3 grid with project breakpoints, collapsing navbar, off-canvas admin menu, scrollable tables	42 browser checks at 1440x900, 768x1024 and Pixel 7 with no horizontal overflow
Accessibility	UI-06, VAL-03	Landmarks, skip link, labels, alt text, focus styles, reduced-motion support, AA contrast	axe-core WCAG 2.2 A/A scans of over 30 page views at three viewports report no violations outstanding (TESTING.md §7)
Entity Relationship Diagram	DB-01, DB-02, DB-04, DB-05, DB-10	Mermaid ERD with every entity, relationship, delete rule, key and index	ERD.md , checked against the migration

Requirement area	IDs	How it is met	Verification
Deployment	DEP-01-03, CI-01, CI-02, PAGES-01, PAGES-02	Docker image + <code>railway.json</code> + <code>/data</code> volume for the application; <code>FirebaseLanding/</code> on Firebase Hosting; three CI jobs	The <code>container</code> and <code>e2e</code> CI jobs; https://learnhub-wapp.web.app . The Railway depl

Technology and platform

ID	Requirement	Type	Implementation	Evidence
TECH-01	Built with .NET technologies – ASP.NET Core MVC on the latest stable LTS (.NET 10)	M	<code>LearnHub.csproj</code> with <code><TargetFramework>net10.0</TargetFramework></code> and <code>Program.cs</code> ; package versions pinned in <code>Directory.Build.props</code> / <code>Directory.Packages.props</code> (ASP.NET Core 10.0.12, EF Core 10.0.12)	CI job <code>build-and-test</code> ; <code>dotnet test</code> locally
TECH-02	C# for all server-side code	M	Every file in <code>Controllers/</code> , <code>Areas/Admin/Controllers/</code> , <code>Services/</code> , <code>Models/</code> , <code>ViewModels/</code> , <code>Infrastructure/</code> and <code>Data/</code> is C#; <code>LangVersion</code> is <code>latest</code>	Repository; build
TECH-03	Database connectivity through Entity Framework Core	M	<code>Data/ApplicationDbContext.cs</code> plus 23 classes under <code>Services/</code> that query through EF Core LINQ; <code>Services/CourseProjections.cs</code> keeps the catalogue queries projected	<code>tests/LearnHub.Tests/Data/DatabaseModelTests.cs</code> ; the integration tests r a real SQLite database
TECH-04	One database provider (SQLite) with code-first EF Core migrations	M	<code>Data/DatabaseServiceCollectionExtensions.cs</code> calls <code>UseSqlite</code> unconditionally and resolves the connection string from <code>DATABASE_CONNECTION_STRING</code> , then <code>ConnectionStrings:DefaultConnection</code> ; one migration folder, <code>Data/Migrations/</code>	<code>Data/DatabaseServiceCollectionExtensions.cs</code> ; CI job <code>container</code> applic migration on an empty volume
TECH-05	ASP.NET Core Identity for authentication	M	<code>Infrastructure/ServiceCollectionExtensions.cs</code> <code>AddLearnHubIdentity</code> with <code>ApplicationUser</code> , <code>IdentityRole</code> , hashing, lockout and cookie configuration	<code>tests/LearnHub.Tests/Integration/AuthenticationTests.cs</code> (9 tests)
TECH-06	Razor views, HTML5 semantic markup	M	72 <code>.cshtml</code> files under <code>Views/</code> and <code>Areas/Admin/Views/</code> ; <code>Views/Shared/_Layout.cshtml</code> uses <code><header></code> , <code><nav></code> , <code><main></code> , <code><footer></code> and a skip link	axe-core scans in <code>tests/e2e/specs/accessibility.spec.js</code> ; <code>Views/Shared/_Layout.cshtml</code>
TECH-07	CSS3 with Bootstrap 5.3 customised by a project design system	M	Bootstrap 5.3.8 in <code>wwwroot/lib/</code> restyled by <code>wwwroot/css/site.css</code> (about 2,800 lines) and <code>wwwroot/css/admin.css</code>	Both stylesheets, served through <code>UseLearnHubStaticFiles</code>
TECH-08	JavaScript for client-side processing where appropriate	M	Progressive enhancement in <code>wwwroot/js/site.js</code> (password-strength meter, quiz timer and counter, confirmations) and <code>wwwroot/js/validation.js</code> (custom adapters)	<code>tests/e2e/specs/public.spec.js</code> registration and contact tests; <code>tests/e2e/specs/student.spec.js</code> quiz test
TECH-09	Multimedia (images, embedded video, PDF documents, SVG illustrations)	M	12 SVG course covers in <code>wwwroot/images/courses/</code> ; embedded YouTube/Vimeo lessons rendered by <code>Services/Content/VideoEmbedParser.cs</code> and <code>Views/Resources/Details.cshtml</code> ; PDF cheat sheets; a data-driven SVG interchange map (<code>Infrastructure/InterchangeMap.cs</code> , <code>Views/Home/Index.cshtml</code>)	<code>AdminManagementTests.Resources_are_created_validated_streamed_and_ContentSafetyTests</code> ; <code>PublicSiteTests.Home_page_map_draws_one_line_per_category</code>

ID	Requirement	Type	Implementation	Evidence
TECH-10	Server-side processing (controllers, services, EF Core queries)	M	9 controllers in <code>Controllers/</code> and 10 in <code>Areas/Admin/Controllers/</code> (plus <code>AdminControllerBase</code>); 23 service classes; no business rule lives in JavaScript	<code>tests/LearnHub.Tests/Services/</code> (22 tests); <code>tests/LearnHub.Tests/Inte</code>
TECH-11	Proper project and file organisation	M	Top level: <code>src/</code> , <code>tests/</code> (<code>LearnHub.Tests</code> , <code>e2e</code>), <code>docs/</code> , <code>scripts/</code> , <code>FirebaseLanding/</code> , <code>.github/workflows/</code> ; inside the project a layered split of <code>Controllers/</code> , <code>Areas/</code> , <code>Services/</code> , <code>Models/</code> , <code>ViewModels/</code> , <code>Infrastructure/</code> , <code>Data/</code> , <code>Views/</code>	Repository tree; ARCHITECTURE.md §2

Guest

ID	Requirement	Type	Implementation	Evidence
GUEST-01	View Home page	M	<code>Controllers/HomeController.cs</code> <code>Index</code> , <code>Views/Home/Index.cshtml</code> (hero, interchange map, featured courses)	<code>PublicSiteTests.Public_pages_load_for_guests</code> ; <code>tests/e2e/specs/public</code> presents the catalogue"
GUEST-02	View About page	M	<code>HomeController.About</code> , <code>Views/Home/About.cshtml</code> ; routed at <code>/About</code> by <code>MapLearnHubRoutes</code>	<code>PublicSiteTests.Public_pages_load_for_guests</code> ; <code>public.spec.js</code> "about p
GUEST-03	Browse published courses	M	<code>Controllers/CoursesController.cs</code> <code>Index</code> , <code>Services/CourseCatalogService.cs</code> (published only)	<code>PublicSiteTests.Seeded_catalogue_is_shown_with_published_courses_only</code> ; <code>Draft_courses_are_not_visible_to_guests</code>
GUEST-04	Search courses (server-side)	M	Server-side <code>EF.Functions.Like</code> over title, description and lesson titles, with wildcards escaped by <code>Services/SearchPattern.cs</code>	<code>PublicSiteTests.Search_is_performed_on_the_server</code> , <code>Search_matches_lesson_titles_and_shows_the_matching_lessons</code> , <code>Search_matches_full_course_descriptions</code> ; <code>ContentSafetyTests.Wildcards_typed_by_users_are_escaped</code>
GUEST-05	Filter courses by category and difficulty	M	<code>CourseSearchQuery</code> filters by category and difficulty with sorting and paging (<code>ViewModels/Public/PublicViewModels.cs</code> , <code>ViewModels/Shared/PagedResult.cs</code>)	<code>PublicSiteTests.Category_filter_limits_results_to_that_category</code> , <code>Courses_beyond_the_first_page_are_reachable_by_paging</code> ; <code>CatalogueServiceTests.Search_returns_only_published_courses_matching</code>
GUEST-06	Open course details	M	<code>CoursesController.Details</code> with the <code>ViewModels/Learning/LearningViewModels.cs</code> course route	<code>PublicSiteTests.Course_details_show_route_and_enrolment_call_to_action</code> "course details and a free preview lesson"
GUEST-07	View selected public (preview) learning resources	M	<code>Services/LearningResourceService.cs</code> releases resources with <code>IsPreview</code> ; <code>Views/Resources/Details.cshtml</code>	<code>PublicSiteTests.Preview_lessons_are_open_to_guests_but_other_lessons,</code>
GUEST-08	Register	M	<code>Controllers/AccountController.cs</code> <code>Register</code> (GET/POST) with <code>ViewModels/Account/AccountViewModels.cs</code>	<code>AuthenticationTests.Registration_creates_a_student_signs_them_in_and</code> , <code>Invalid_registration_is_rejected_by_server_side_validation</code> , <code>Registering_with_an_existing_email_shows_a_single_clear_error</code>
GUEST-09	Login	M	<code>AccountController.Login</code> (GET/POST) with a local-return-URL guard	<code>AuthenticationTests.Students_and_admins_are_sent_to_their_own_dashboards</code> , <code>Wrong_password_shows_a_generic_error</code> , <code>Login_redirects_back_to_a_local_return_url_but_never_to_another_site</code>
GUEST-10	View Contact page and send a contact message	M	<code>HomeController.Contact</code> (GET/POST), <code>Services/ContactService.cs</code> stores the message and discards honeypot submissions; the POST is rate limited	<code>AdminManagementTests.Contact_messages_reach_the_admin_inbox</code> , <code>Spam_honeypot_submissions_are_not_stored</code> ; <code>public.spec.js</code> "contact form before submitting"

Student

ID	Requirement	Type	Implementation	Evidence
STU-01	Student dashboard with meaningful, data-driven information	M	<code>Controllers/StudentController.cs</code> <code>Dashboard</code> , <code>Services/DashboardService.cs</code> , <code>Views/Student/Dashboard.cshtml</code>	<code>tests/e2e/specs/student.spec.js</code> "dashboard, course progress, lesson and quiz"

ID	Requirement	Type	Implementation	Evidence
STU-02	Dashboard: enrolled courses, completed courses, available courses	M	Dashboard figures for enrolled, completed and available courses (ViewModels/Learning/LearningViewModels.cs)	Student/Dashboard.cshtml ; CatalogueServiceTests.Progress_counts_completed_lessons_and_passed_quiz
STU-03	Dashboard: recent activity, quiz results, learning progress	M	Recent activity feed, recent quiz results, progress bars and recommendations from Services/ProgressService.cs	Student/Dashboard.cshtml and Views/Student/Progress.cshtml ; CatalogueServiceTests.Progress_counts_completed_lessons_and_passed_quiz Stored_progress_is_recalculated_and_completion_is_cleared_when_new_course
STU-04	View and update profile	M	Controllers/ProfileController.cs Index (GET/POST), including a guarded e-mail change	StudentJourneyTests.Profile_updates_are_validated_and_saved , Changing_the_email_address_needs_the_current_password_and_an_unused_ac
STU-05	Browse courses and view course details	M	The catalogue and details pages accept anonymous and authenticated visitors	NavigationCrawlTests.Student_navigation_has_no_broken_links ; student.
STU-06	Enrol in a course (and leave a course)	M	CoursesController.Enroll / Leave (POST, [Authorize(Roles = "Student")]), Services/EnrollmentService.cs	CatalogueServiceTests.Enrolling_twice_is_rejected_and_unpublished_course Leaving_a_course_keeps_completed_lessons_for_later ; StudentJourneyTests.Leaving_a_course_removes_access_to_its_lessons
STU-07	View My Courses	M	StudentController.MyCourses with an MyCoursesFilter (all, in progress, completed)	Student/MyCourses.cshtml ; student.spec.js
STU-08	Access learning resources of enrolled courses	M	ResourcesController.Details and Open (stream/download) enforce enrolment in Services/LearningResourceService.cs ; files are streamed from outside wwwroot	StudentJourneyTests.A_new_student_enrols_studies_takes_the_quiz_and_se PublicSiteTests.Private_files_cannot_be_downloaded_by_guests
STU-09	Track learning progress (mark resources complete)	M	ResourcesController.ToggleComplete , Services/ProgressService.cs recalculates the enrolment percentage	CatalogueServiceTests.Progress_counts_completed_lessons_and_passed_quiz Draft_lessons_are_hidden_from_students_and_do_not_count_towards_progre
STU-10	Take quizzes	M	Controllers/QuizzesController.cs Index / Take / Submit , Services/QuizService.cs and Services/QuizGrader.cs ; the page never contains the correct answers	QuizAndCourseManagementTests.Submitting_answers_stores_a_graded_attempt The_quiz_page_never_contains_the_correct_answers , Students_who_are_not_enrolled_cannot_take_or_submit_the_quiz ; LogicTe
STU-11	View quiz results and history	M	QuizzesController.Result and History , Views/Quizzes/Result.cshtml with _QuizReview.cshtml explanations	QuizAndCourseManagementTests.Students_cannot_open_other_students_result student.spec.js
STU-12	Logout	M	POST-only AccountController.Logout with [Authorize] , so a link cannot sign a user out	AuthenticationTests.Logout_ends_the_session
STU-13	Change password	O	ProfileController.ChangePassword (GET/POST), rate limited, refreshes the sign-in cookie after a change	ProfileController.cs ; axe-core scan of /Profile/ChangePassword in accessi

Administrator

ID	Requirement	Type	Implementation	Evidence
ADM-01	Separate, protected Admin area	M	Areas/Admin/ with its own layout; Areas/Admin/Controllers/AdminControllerBase.cs carries [Authorize(Roles = AppRoles.Admin)] , so every derived controller inherits it	AuthorizationTests.Guests_are_sent_to_login_for_admin_pages Students_are_denied_access_to_admin_pages , Students_cannot_post_to_admin_actions_even_with_a_valid_token
ADM-02	Admin dashboard with real platform statistics	M	Areas/Admin/Controllers/DashboardController.cs and Services/DashboardService.cs ; Areas/Admin/Views/Dashboard/Index.cshtml	tests/e2e/specs/admin.spec.js "dashboard and management pages"
ADM-03	Courses CRUD (create, list, details, edit, delete)	M	Admin CoursesController (index, details, create, edit, TogglePublished , delete/confirm) with Services/CourseManagementService.cs	AdminManagementTests.Course_create_with_cover_edit_publish_and Category_with_courses_is_protected_from_deletion

ID	Requirement	Type	Implementation	Evidence
ADM-04	Categories CRUD	M	Admin <code>CategoriesController</code> , <code>Services/CategoryService.cs</code> (unique name, protected delete)	<code>AdminManagementTests.Category_create_edit_and_delete</code> , <code>Category_validation_errors_are_shown</code> ; <code>CatalogueServiceTests.Creates_categories_and_rejects_duplicates</code> ; <code>A_category_that_still_has_courses_cannot_be_deleted</code> ; admin delete through the forms"
ADM-05	Learning resources CRUD (incl. secure file upload)	M	Admin <code>ResourcesController</code> , <code>Services/ResourceManagementService.cs</code> , <code>Services/Storage/FileStorageService.cs</code> and <code>Infrastructure/FileValidationAttributes.cs</code>	<code>AdminManagementTests.Resources_are_created_validated_streamed</code> , <code>FileStorageServiceTests</code> (7 tests)
ADM-06	Enrolments management (list, enrol a student, remove)	M	Admin <code>EnrollmentsController</code> (index, create, delete/confirm) with <code>Services/EnrollmentService.cs</code>	<code>AdminManagementTests.Enrolments_can_be_managed_by_administrator</code>
ADM-07	Quizzes CRUD	M	Admin <code>QuizzesController</code> (index, details, create, edit, <code>SetPublished</code> , delete) with <code>Services/QuizManagementService.cs</code>	<code>AdminManagementTests.Quiz_must_have_questions_before_it_can_be_created</code>
ADM-08	Quiz questions and answer options CRUD, set correct answer	M	Admin <code>QuestionsController</code> with <code>ViewModels/Admin/AdminLearningViewModels.cs</code> ; <code>Services/QuestionManagementService.cs</code> enforces 2–6 options with exactly one correct answer	<code>LogicTests.Question_needs_two_answers_one_marked_correct_and_one_wrong</code> ; <code>QuizAndCourseManagementTests.Option_ids_that_belong_to_another_quiz_cannot_be_reused</code>
ADM-09	Quiz results (attempts) view and delete	M	Admin <code>QuizAttemptsController</code> (index with filters, details, delete/confirm)	Link crawl as administrator (<code>NavigationCrawlTests.Admin_navigation_scan_of_Admin_QuizAttempts</code> — there is no dedicated automated test for this)
ADM-10	Users: list, view details, change role, deactivate/reactivate, safe delete	M	Admin <code>UsersController</code> (index, details, <code>SetRole</code> , <code>Deactivate</code> , <code>Reactivate</code> , delete/confirm) with <code>Services/UserManagementService.cs</code> , which protects the last administrator and self-deletion	<code>AdminManagementTests.User_roles_deactivation_and_self_protection</code>
ADM-11	Website content: contact message inbox	O	Admin <code>MessagesController</code> (index, details, <code>MarkUnread</code> , delete) with <code>Services/ContactService.cs</code>	<code>AdminManagementTests.Contact_messages_reach_the_admin_inbox</code>

Database

ID	Requirement	Type	Implementation	Evidence
DB-01	Relational model with primary keys and foreign keys	M	Fluent API configuration in <code>Data/Configurations/UserAndCatalogConfigurations.cs</code> and <code>Data/Configurations/QuizConfigurations.cs</code> over the 11 application entities in <code>Models/</code>	<code>DatabaseModelTests.Create_script_enforces_the_integrity_rules</code>
DB-02	Required properties and maximum lengths	M	<code>IsRequired()</code> and <code>HasMaxLength()</code> taken from <code>Models/FieldLengths.cs</code> , which the view models reuse so form rules cannot drift from the schema	<code>DatabaseModelTests.Create_script_enforces_the_integrity_rules</code>
DB-03	Unique constraints where appropriate	M	Unique indexes on category name, enrolment (user + course), resource completion (user + resource) and quiz answer (attempt + question)	<code>DatabaseModelTests.Create_script_enforces_the_integrity_rules</code> ; <code>CatalogueServiceTests.Creates_categories_and_rejects_duplicates</code>
DB-04	Indexes where useful	M	Indexes on search, filter, ordering and foreign-key columns, for example <code>Courses(IsPublished, CategoryId)</code> , <code>Courses(Title)</code> , <code>LearningResources(CourseId, SortOrder)</code> , <code>Enrollments(CourseId, CompletionPercentage)</code> , <code>QuizAttempts(CompletedAt)</code>	Index table in ERD.md ; 27 <code>CreateIndex</code> calls in the migration
DB-05	Intentional cascade / restrict delete behaviour	M	Restrict for <code>Course → Category</code> and both <code>QuizAnswer</code> links; cascade elsewhere; <code>Data/TransactionExtensions.cs</code> runs the ordered deletes in one transaction	<code>DatabaseModelTests.Delete_behaviour_is_intentional</code> ; <code>QuizAndCourseManagementTests.Deleting_a_course_removes_its_content</code>
DB-06	CreatedAt / UpdatedAt timestamps where useful	M	<code>Models/IHasTimestamps.cs</code> is applied in <code>ApplicationDbContext.SaveChanges</code> ; <code>Data/UtcDateTimeConverter.cs</code> stores UTC	<code>ApplicationDbContext.cs</code> ; ERD.md

ID	Requirement	Type	Implementation	Evidence
DB-07	EF Core migrations reproducible from an empty database	M	Single migration <code>Data/Migrations/20260914100832_InitialCreate.cs</code> creates 18 tables (11 application + 7 Identity); applied at start-up when <code>Database:ApplyMigrationsOnStartup</code> is true; <code>Data/DesignTimeDbContextFactory.cs</code> is the one design-time factory	The <code>container</code> CI job creates the schema on an empty volume, then restarts <code>tests/LearnHub.Tests/Infrastructure/TestDatabase.cs</code> migrates a f
DB-08	Realistic seed data (categories, courses, resources, quizzes, demo activity)	M	<code>Data/Seed/DatabaseInitializer.cs</code> → <code>DemoDataSeeder</code> → <code>DemoCatalog</code> (four partial files); seeded content measured: 8 categories, 12 courses (11 published, 1 draft), 64 resources, 11 quizzes, 54 questions, 216 answer options, 7 learners	The <code>container</code> CI job asserts the log line "Demo data seeded"; counts verified
DB-09	Insert, Display, Update, Delete operations demonstrated	M	Insert, display, update and delete are exercised through the student and admin screens rather than raw SQL	<code>AdminManagementTests</code> (10 tests) and <code>StudentJourneyTests</code> (5 tests) dr
DB-10	ERD that exactly matches the EF Core entities	M	ERD.md generated from <code>Models/</code> and <code>Data/Configurations/</code>	Entity, key, index and delete-behaviour tables checked against <code>Data/Migratio</code>

Authentication and authorisation

ID	Requirement	Type	Implementation	Evidence
AUTH-01	Register, Login, Logout	M	<code>Controllers/AccountController.cs</code> plus <code>Views/Account/Register.cshtml</code> and <code>Login.cshtml</code>	<code>AuthenticationTests</code> (9 tests); <code>public.spec.js</code> registration test
AUTH-02	Password hashing through Identity (no plain-text passwords)	M	<code>AddIdentity</code> with the default <code>PasswordHasher</code> ; no password field is ever stored or logged	<code>AuthenticationTests.Registration_creates_a_student_signs_them_</code>
AUTH-03	Roles: Guest, Student, Admin with role-based authorisation	M	<code>Models/AppRoles.cs</code> creates <code>Admin</code> and <code>Student</code> ; anonymous visitors are the guest role; <code>Infrastructure/LearnHubClaimsPrincipalFactory.cs</code> adds the role claim	<code>DatabaseInitializer.EnsureRolesAsync</code> ; <code>AuthorizationTests</code> (8 test
AUTH-04	Protected pages for students and admin-only controllers	M	<code>[Authorize(Roles = AppRoles.Student)]</code> on <code>StudentController</code> , <code>QuizzesController</code> and the student-only POST actions; <code>AdminControllerBase</code> covers the Admin area	<code>AuthorizationTests.Students_can_open_their_own_pages</code> , <code>Administrators_are_denied_student_only_pages</code> , <code>Access_denied_pa</code> <code>NavigationCrawlTests</code> crawls each role separately
AUTH-05	Initial admin seeded from secrets / environment variables (no password in Git)	M	<code>Data/Seed/DatabaseInitializer.cs</code> creates the administrator from <code>Seed:AdminEmail / Seed:AdminPassword</code> ; <code>appsettings.json</code> ships both passwords empty; <code>.env.example</code> documents them with placeholders only; in Development <code>DevelopmentSeedPasswords</code> writes generated passwords to the git-ignored <code>App_Data/demo-credentials.json</code>	<code>DevelopmentSeedPasswordsTests</code> (4 tests); the <code>container</code> CI job generat and asserts the log line "Administrator account ... created"
AUTH-06	Account lockout after repeated failed logins	O	<code>AddIdentity</code> lockout: 5 failed attempts, 15-minute lockout, enabled for new users; the failed attempt is logged	<code>AuthenticationTests.Account_is_locked_after_five_failed_attempt</code>

Validation

ID	Requirement	Type	Implementation	Evidence
VAL-01	Server-side validation with <code>DataAnnotations</code> and <code>ModelState</code>	M	<code>DataAnnotations</code> on every view model in <code>ViewModels/</code> ; controllers re-check <code>ModelState.IsValid</code> and services return <code>Services/OperationResult.cs</code>	<code>AuthenticationTests.Invalid_registration_is_rejected_by_server_si</code> <code>AdminManagementTests.Category_validation_errors_are_shown</code>

ID	Requirement	Type	Implementation	Evidence
VAL-02	Client-side validation with ASP.NET validation helpers	M	jQuery Validation Unobtrusive plus custom adapters in <code>wwwroot/js/validation.js</code> ; <code>Views/Shared/_ValidationScriptsPartial.cshtml</code>	<code>public.spec.js</code> "contact form validates in the browser before submitting" and "re gives live password feedback"
VAL-03	Bootstrap-styled, accessible validation feedback	M	Validation messages rendered with Bootstrap's <code>is-invalid</code> / <code>invalid-feedback</code> , <code>aria-describedby</code> hints and <code>aria-live</code> status regions	axe-core scans in <code>accessibility.spec.js</code> ; <code>Views/Shared/_Layout.cshtml</code>
VAL-04	Required, email, password, length and range rules	M	<code>[Required]</code> , <code>[EmailAddress]</code> , <code>[StringLength]</code> (from <code>Models/FieldLengths.cs</code>) , <code>[Range]</code> , <code>[Compare]</code> and the project's <code>MustBeTrueAttribute</code>	<code>LogicTests.Register_requires_matching_strong_passwords_and_accepts_a_valid_registration_has_no_errors</code> ; <code>LogicTests.Resource_rules_depend_on_the_resource_type</code>
VAL-05	File validation (type, size, content signature)	M	<code>Infrastructure/FileValidationAttributes.cs</code> (<code>[MaxFileSize]</code> , <code>[AllowedExtensions]</code>) plus signature and real-length checks in <code>Services/Storage/FileStorageService.cs</code> with limits in <code>Services/Storage/UploadRules.cs</code>	<code>FileStorageServiceTests</code> (7 tests)
VAL-06	Business-rule validation (e.g. exactly one correct answer)	M	Service rules: exactly one correct answer and 2–6 options, a quiz needs questions before publishing, unique category names, an e-mail change needs the current password, and the last administrator cannot be removed	<code>LogicTests.Question_needs_two_answers_one_marked_correct_and_no_duplicates</code> ; <code>AdminManagementTests.Quiz_must_have_questions_before_it_can_be_published</code> ; <code>User_roles_deactivation_and_self_protection</code> ; <code>CatalogueServiceTests</code>

Security

ID	Requirement	Type	Implementation	Evidence
SEC-01	No SQL injection (EF Core parameterised queries only)	M	EF Core LINQ everywhere; the only raw pattern is an escaped <code>LIKE</code> term built by <code>Services/SearchPattern.cs</code> , and no SQL string is concatenated	<code>ContentSafetyTests.Wildcards_typed_by_users_are_escaped</code> , <code>Very_long_search_terms_are_truncated</code> ; SECURITY.md
SEC-02	XSS protection (Razor encoding, no raw user HTML, CSP)	M	Razor encodes by default; the only markup renderer, <code>Services/Content/LessonContentRenderer.cs</code> , emits allow-listed elements; <code>Infrastructure/SecurityHeadersMiddleware.cs</code> sends a strict CSP with <code>script-src 'self'</code>	<code>ContentSafetyTests.Html_in_lesson_text_is_encoded_and_never_executed</code> ; <code>Formatting_syntax_produces_only_allow_listed_elements</code> , <code>Attack_attempts_through_formatting_markers_stay_as_text</code> ; <code>PublicSiteTests.Html_responses_carry_security_headers</code>
SEC-03	CSRF protection (anti-forgery tokens on all unsafe requests)	M	Global <code>AutoValidateAntiforgeryTokenAttribute</code> registered in <code>Infrastructure/ServiceCollectionExtensions.cs</code> ; every POST form emits a token	<code>AuthenticationTests.Posts_without_an_anti_forgery_token_are_rejected</code>
SEC-04	IDOR protection (ownership checks on attempts, resources, files)	M	Ownership and enrolment checks in <code>Services/QuizService.cs</code> , <code>LearningResourceService.cs</code> and <code>ResourceManagementService.cs</code> ; files are streamed through a controller action, never served from <code>wwwroot</code>	<code>QuizAndCourseManagementTests.Students_cannot_open_other_students_resources</code> ; <code>Students_who_are_not_enrolled_cannot_take_or_submit_the_quiz</code> ; <code>PublicSiteTests.Private_files_cannot_be_downloaded_by_guests</code>
SEC-05	Overposting protection (ViewModels, never bind entities)	M	Every action binds a view model from <code>ViewModels/</code> ; no <code>[Bind]</code> /entity binding anywhere	<code>StudentJourneyTests.Tampered_quiz_answers_are_graded_on_the_server</code> ; <code>QuizAndCourseManagementTests.Option_ids_that_belong_to_another_quiz_cannot_be_selected</code>
SEC-06	Safe file uploads (allow-list, magic bytes, size limit, random names, outside <code>wwwroot</code>)	M	Allow-list of extensions and MIME types, content-signature check, size limit, GUID file names, storage under <code>Storage:RootPath</code> (on Railway <code>/data/storage</code>), and path-traversal rejection	<code>FileStorageServiceTests</code> — including <code>Rejects_a_file_whose_content_does_not_match_its_extension</code> , <code>Rejects_files_over_the_size_limit_even_if_the_declared_length_is_within_limit</code> , <code>Paths_outside_the_storage_root_are_never_resolved</code>

ID	Requirement	Type	Implementation	Evidence
SEC-07	Safe video embedding (strict YouTube/Vimeo parsing, frame-src allow-list)	M	<code>Services/Content/VideoEmbedParser.cs</code> accepts only well-formed HTTPS YouTube/Vimeo links and builds a privacy-enhanced embed; CSP <code>frame-src</code> allows only <code>www.youtube-nocookie.com</code> and <code>player.vimeo.com</code>	<code>ContentSafetyTests</code> video cases (19 of the 21 inline data rows in that theory)
SEC-08	No secrets committed to Git	M	Passwords and connection strings come from environment variables (<code>DATABASE_CONNECTION_STRING</code> , <code>Seed__AdminPassword</code> , <code>Seed__DemoStudentPassword</code> , <code>DataProtection__KeysPath</code>); <code>.env</code> is git-ignored and only <code>.env.example</code> with placeholders is committed; CI generates throw-away passwords per run and masks them; the development credential file is git-ignored	<code>.gitignore</code> ; <code>.env.example</code> ; <code>SECURITY.md</code> §3
SEC-09	Admin routes protected by role authorisation	M	<code>AdminControllerBase</code> <code>[Authorize(Roles = AppRoles.Admin)]</code> is inherited by all 10 admin controllers	<code>AuthorizationTests.Students_are_denied_access_to_admin_pages</code> , <code>Students_cannot_post_to_admin_actions_even_with_a_valid_token</code>
SEC-10	Security headers, HTTPS/HSTS in production	O	<code>SecurityHeadersMiddleware</code> adds CSP, X-Content-Type-Options, X-Frame-Options, Referrer-Policy, Permissions-Policy and COOP; <code>Program.cs</code> adds HSTS and HTTPS redirection outside Development; <code>Infrastructure/ApplicationBuilderExtensions.cs</code> <code>UsePlatformProxyHeaders</code> reads X-Forwarded-Proto so the scheme is correct behind the Railway proxy	<code>PublicSiteTests.Html_responses_carry_security_headers</code> ; <code>scripts/s</code> headers on a running instance
SEC-11	Rate limiting on login, registration and contact form	O	Fixed-window limiter (10 requests per 60 s per client IP, <code>RateLimiting:PermitLimit</code>) applied by <code>[EnableRateLimiting(RateLimitPolicies.Forms)]</code> to registration, login, contact, profile update and password change	<code>Infrastructure/ServiceCollectionExtensions.cs</code> and the four attributed 429

Error handling and logging

ID	Requirement	Type	Implementation	Evidence
ERR-01	Custom 404 page	M	<code>Program.cs</code> <code>UseStatusCodePagesWithReExecute("/Error/{0}")</code> → <code>Controllers/ErrorController.cs</code> <code>Status</code> → <code>Views/Error/Status.cshtml</code> (branded, with a support reference)	<code>PublicSiteTests.Unknown_pages_and_invalid_ids_return_a_friendly_404</code> "unknown pages show the friendly 404 page"; <code>NavigationCrawlTests</code> fails on any
ERR-02	Custom 500 / error page without stack traces in production	M	<code>UseExceptionHandler("/Error")</code> outside Development; the friendly page never renders an exception, only a request id	The <code>container</code> CI job runs Production mode and fails on any <code>fail:</code> / <code>crit:</code> log I <code>ErrorController.ServerError</code>
ERR-03	User-friendly validation and business error messages	M	<code>Services/OperationResult.cs</code> messages surface through <code>TempData</code> (<code>Infrastructure/StatusMessages.cs</code>) and <code>ModelState</code> , plus empty states in the views	<code>AdminManagementTests.Category_validation_errors_are_shown</code> , <code>Quiz_must_have_questions_before_it_can_be_published</code> ; <code>CatalogueServ</code>
ERR-04	Database error handling (e.g. unique-constraint races)	M	<code>DbUpdateException</code> from a unique index is caught and turned into a friendly message in <code>Services/EnrollmentService.cs</code> , <code>CategoryService.cs</code> and <code>ResourceManagementService.cs</code>	<code>CatalogueServiceTests.Enrolling_twice_is_rejected_and_unpublished_c</code> <code>Creates_categories_and_rejects_duplicate_names_regardless_of_case</code>
ERR-05	Structured logging of important events	M	Structured <code>ILogger</code> events for start-up, migrations, seeding, sign-in, logout, enrolment, quiz submission, deletion, upload and honeypot rejection	The <code>container</code> CI job asserts the migration, seeding and administrator log lines and <code>Data/Seed/DatabaseInitializer.cs</code>

UI/UX, accessibility and responsiveness

ID	Requirement	Type	Implementation	Evidence	Status
UI-01	Coherent custom design system (not default Bootstrap)	M	The Interchange design system in <code>wwwroot/css/site.css</code> (colour, type, spacing and component tokens) plus <code>wwwroot/css/admin.css</code> ; the same tokens are reused by <code>FirebaseLanding/assets/site.css</code>	Both stylesheets; https://learnhub-wapp.web.app	PA
UI-02	Navigation bar per role, hero, course cards, dashboards, tables, forms, footer	M	<code>Views/Shared/_Layout.cshtml</code> renders a different navigation per role; <code>Views/Shared/_CourseCard.cshtml</code> , <code>_ProgressRoute.cshtml</code> , <code>_Pagination.cshtml</code> , <code>_StatusMessage.cshtml</code> ; <code>Areas/Admin/Views/Shared/_AdminLayout.cshtml</code> has an off-canvas sidebar	All public, student and admin screenshots; <code>admin.spec.js</code> , <code>student.spec.js</code>	PA
UI-03	Empty states and success/error messages	M	Empty states in the catalogue, My Courses, dashboard, progress and admin lists; flash messages through <code>Infrastructure/StatusMessages.cs</code> and <code>Views/Shared/_StatusMessage.cshtml</code> with <code>aria-live</code>	The views above; <code>AdminManagementTests</code> assert the resulting messages are shown	PA
UI-04	Breadcrumbs where useful	M	Breadcrumb trails on nested pages — course details, lesson, quiz take/result, profile and the admin detail/form pages	20 views contain a breadcrumb; NAVIGATION.md §3	PA
UI-05	Responsive on mobile, tablet and desktop; collapsing navigation; usable tables	M	Bootstrap 5.3 grid plus project breakpoints; collapsing navbar, off-canvas admin menu, and <code>.table-wrap</code> containers that scroll horizontally and stack into label/value rows on phones (<code>table-stack</code>)	42 browser checks (14 tests × desktop 1440×900, tablet 768×1024, mobile Pixel 7) assert no horizontal overflow; <code>public.spec.js</code> "navigation menu works on every screen size"	PA
UI-06	Semantic HTML, labels, alt text, keyboard access, contrast, heading structure	M	Landmarks and a skip link in <code>_Layout.cshtml</code> , explicit labels and <code>aria-describedby</code> hints, decorative images <code>aria-hidden</code> with empty <code>alt</code> , visible <code>:focus-visible</code> styles, and <code>prefers-reduced-motion</code> handling	axe-core WCAG 2.2 A/A+ scans of the public, student and admin pages at all three viewports report no violations; Keyboard and screen-reader passes are still listed as "To do" in TESTING.md §7	PA
UI-07	No placeholder/template text, no broken images or dead links	M	Every seeded course, lesson, quiz and image is real content; no lorem ipsum and no template placeholders remain	<code>NavigationCrawlTests</code> crawls all three roles with no broken links; <code>PublicSiteTests.Static_assets_and_seeded_covers_are_served</code> ; the browser tests fail on console errors and broken images. The four names in the team table of PROPOSAL.md and TEAM.md are intentional placeholders.	PA

Testing and quality

ID	Requirement	Type	Implementation	Evidence	Status
TEST-01	Automated unit tests for services and helpers	M	<code>tests/LearnHub.Tests/Unit/</code> (5 files: logic, content safety, file storage, seed passwords) and <code>tests/LearnHub.Tests/Services/</code> (2 files)	211 tests pass, 0 fail, in <code>dotnet test LearnHub.sln -c Release</code>	PASS
TEST-02	Automated integration tests (auth, protected routes, CRUD, enrolment, quiz, search, validation, 404)	M	<code>tests/LearnHub.Tests/Integration/</code> — <code>PublicSiteTests</code> , <code>AuthenticationTests</code> , <code>AuthorizationTests</code> , <code>AdminManagementTests</code> , <code>StudentJourneyTests</code> , <code>NavigationCrawlTests</code> ; <code>tests/LearnHub.Tests/Infrastructure/LearnHubWebApplicationFactory.cs</code> runs the real pipeline in the <code>Testing</code> environment, and <code>tests/LearnHub.Tests/Infrastructure/TestDatabase.cs</code> gives each test its own migrated SQLite database	Same run: 211 pass, 0 fail	PASS
TEST-03	Browser end-to-end tests across mobile, tablet and desktop viewports	O	<code>tests/e2e/specs/</code> — <code>public</code> , <code>student</code> , <code>admin</code> and <code>accessibility</code> specs; <code>tests/e2e/playwright.config.js</code> defines the desktop, tablet and mobile projects; the <code>e2e</code> CI job serves a Production instance over HTTPS	42/42 browser checks pass (14 tests × 3 viewports)	PASS
TEST-04	Manual testing checklist with recorded results	M	TESTING.md §7 "Manual testing checklist" records 13 results	13 items passed. Four remain "To do" and need a person or a real deployment: keyboard-only journeys, a screen-reader pass, real devices, and the live production check	PASS

ID	Requirement	Type	Implementation	Evidence	Status
TEST-05	Build without errors and without warnings	M	The <code>build-and-test</code> CI job runs <code>dotnet build LearnHub.sln -c Release -p:TreatWarningsAsErrors=true</code> ; <code>Directory.Build.props</code> keeps package-vulnerability advisories (<code>NU1901</code> – <code>NU1904</code>) visible as warnings without breaking unrelated builds	CI build: 0 warnings, 0 errors	PASS

Repository, CI/CD and deployment

ID	Requirement	Type	Implementation	Evidence	Status
GIT-01	Git repository with a professional structure and <code>.gitignore</code>	M	https://github.com/KhayitOff/LearnHub on branch <code>feature/railway-firebase-sqlite</code> ; <code>.gitignore</code> covers build output, <code>App_Data/</code> , <code>.env*</code> (except <code>.env.example</code>) and generated credentials: <code>.gitattributes</code> normalises line endings	Repository	PASS
GIT-02	Sensible commit history and documented branching strategy	M	Conventional Commits history with scoped subjects; the branching strategy is in GIT_WORKFLOW.md	<code>git log</code> ; GIT_WORKFLOW.md	PASS
GIT-03	README with all required sections	M	<code>README.md</code> — overview, technology, structure, setup, tests and a documentation index	<code>README.md</code> . Its badges, hosting table and deployment section still describe the earlier Azure hosting and are not yet updated.	PASS
CI-01	GitHub Actions CI: restore, build, test, fail on error	M	<code>.github/workflows/ci.yml</code> job <code>build-and-test</code> — checkout, .NET 10, cached restore, <code>dotnet build -p:TreatWarningsAsErrors=true</code> , the whole test suite, a result summary and the TRX artifact; runs on every branch push and on pull requests to <code>main</code> / <code>develop</code> , ignoring <code>docs/</code> , <code>FirebaseLanding/</code> and Markdown-only changes	The job file; the latest CI run	PASS
CI-02	GitHub Actions container workflow that reproduces the Railway runtime	M	<code>.github/workflows/ci.yml</code> job <code>container</code> — builds the Docker image, starts it with Production settings, <code>PORT=8080</code> and a volume at <code>/data</code> , waits for <code>/health</code> , runs <code>scripts/smoke-test.sh</code> , asserts the start-up log shows migrations applied, demo data seeded and the administrator created with no <code>fail: / crit:</code> lines, then restarts the container and asserts migrations are not re-applied	The job file; the uploaded <code>container</code> artifact (both logs)	PASS
DEP-01	Production ASP.NET application packaged and ready to run on Railway	M	<code>Dockerfile</code> (multi-stage <code>sdk:10.0</code> → <code>aspnet:10.0</code>), <code>railway.json</code> (DOCKERFILE builder, health check <code>/health</code> , 300 s timeout, restart on failure), <code>Infrastructure/ApplicationBuilderExtensions.cs</code> <code>UsePlatformPort</code> binding <code>http://0.0.0.0:{PORT}</code> and <code>UsePlatformProxyHeaders</code> for the TLS-terminating proxy, and <code>.env.example</code> documenting every variable	The <code>container</code> CI job runs this image the way Railway does; Railway CLI is unauthenticated, so no Railway deployment has been performed	READY – not deployed
DEP-02	SQLite production database on a persistent volume, configured through environment variables	M	<code>DATABASE_CONNECTION_STRING=Data Source=/data/learnhub.db</code> , <code>DataProtection__KeysPath=/data/keys</code> and <code>Storage__RootPath=/data/storage</code> ; <code>Data/DatabaseServiceCollectionExtensions.cs</code> prefers the environment variable over <code>ConnectionStrings:DefaultConnection</code> and creates relative folders; migrations are applied at start-up	The <code>container</code> CI job mounts a volume at <code>/data</code> , seeds it, restarts the container and proves the SQLite file survived; no Railway volume has been created yet	READY – not deployed
DEP-03	Documented migration strategy and production verification	M	Start-up migrations guarded by <code>Database:ApplyMigrationsOnStartup</code> ; <code>.github/workflows/ef-migrations.yml</code> generates a migration in the cloud and commits it; the health endpoint <code>/health</code> uses <code>Infrastructure/DatabaseHealthCheck.cs</code> ; <code>scripts/smoke-test.sh</code> verifies a running instance	DEPLOYMENT.md ; <code>.github/workflows/ef-migrations.yml</code> ; <code>scripts/smoke-test.sh</code>	PASS
PAGES-01	Separate static presentation site with an "Open Learning System" call to action	M	<code>FirebaseLanding/</code> — <code>index.html</code> , <code>404.html</code> and <code>assets/</code> (design system, Overpass webfont and eight real screenshots); its buttons link to the Railway application, the GitHub repository and the documentation	Live at https://learnhub-wapp.web.app (<code>/ 200</code> , <code>/assets/site.css</code> 200, <code>/assets/screens/home-desktop.png</code> 200, unknown path 404). The buttons carry the conventional Railway address until <code>scripts/set-app-url.sh</code> rewrites it after the first deploy.	PASS
PAGES-02	Presentation site deployment automated and reproducible	M	<code>firebase.json</code> at the repository root sets <code>"public": "FirebaseLanding"</code> with <code>cleanUrls</code> and asset caching headers; <code>.firebaserc</code> sets the default project to <code>learnhub-wapp</code> ; deployment is one command, <code>firebase deploy --only hosting</code>	<code>firebase.json</code> , <code>.firebaserc</code> ; the live site. The former GitHub Pages site and its <code>pages.yml</code> workflow were deleted, so no heading or file named <code>pages.yml</code> remains.	PASS

Documentation

ID	Requirement	Type	Implementation	Evidence	Status
DOC-01	Proposal: title, objectives, mission, audience modelling, scope	M	PROPOSAL.md	File	PASS
DOC-02	Final report content (full structure from the brief)	M	REPORT_NOTES.md	File	PASS
DOC-03	ERD (Mermaid)	M	ERD.md	File	PASS
DOC-04	Use cases	M	USE_CASES.md	File	PASS
DOC-05	Flowcharts (registration, login, enrolment, quiz, admin course CRUD)	M	FLOWCHARTS.md — registration, login, enrolment, lesson study, quiz attempt, admin course CRUD and more	File	PASS
DOC-06	Wireframes	M	WIREFRAMES.md	File	PASS
DOC-07	Navigation structure diagram	M	NAVIGATION.md	File	PASS
DOC-08	Testing documentation	M	TESTING.md	File	PASS
DOC-09	Deployment documentation	M	DEPLOYMENT.md	File	PASS
DOC-10	Team responsibilities and Git workflow	M	TEAM.md , GIT_WORKFLOW.md	Files	PASS
DOC-11	Viva preparation questions	M	VIVA.md	File	PASS